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Paleomagnetism of Cretaceous to lower Tertiary rocks from Southwestern Sichuan: a revisit

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ABSTRACT

A re-study has been carried out on the Cretaceous to lower Tertiary rocks from southwestern Sichuan province on the South China Block. The characteristic remanent magnetization (ChRM) resolved from the Paleocene–Eocene Leidashu Formation is of dual polarity and very shallow inclination. The anomalously low paleolatitude ($6.9 \pm 6.3°$N) indicated by the inclination is not supported by geological evidence from this region and is probably due to inclination error related to depositional and/or post-depositional processes. A positive fold test, as well as the prevalence of normal polarity, suggests that the ChRM of the Mid–Upper Cretaceous Xiaoba Formation might be acquired during the Cretaceous Long Normal Superchron (CLNS). The tilt-corrected mean direction of the Xiaoba Formation ($D/I = 8.1°/38.8°$, $\alpha_{95} = 6.6°$, $N = 18$ sites) is not significantly different from a preliminary result ($D/I = 357.6°/31.6°$, $\alpha_{95} = 8.6°$, $N = 12$ samples) we reported earlier for the collective Xiaoba and Leidashu formations. The resultant paleopole (81.5°N/220.9°E, $A_{95} = 7.1°$) is coincident with other Cretaceous poles for the Chinese blocks. Comparison with the Eurasian apparent polar wander path (APWP) indicates neither significant rotation nor latitudinal translation of the study area relative to Eurasia. The newly obtained mean direction for the Xiaoba Formation is believed to be more reliable than our earlier result since it is based on a larger sample size and is apparently free from effects of the shallow inclination observed in the Leidashu Formation, and should supercede our previous preliminary result. The paleopole derived from seven sites of the Lower Cretaceous Feitianshan Formation (69.0°N/204.6°E, $A_{95} = 4.3°$) deviates from the track from the Late Jurassic to Late Cretaceous poles; the deviation might be reduced by further sampling.

Introduction

The last decade has seen a dramatic increase in high quality paleomagnetic data from China since McElhinny and his co-workers conducted the first paleomagnetic study using modern techniques [1]. There is a general consensus among paleomagnetists working in China that, by Cretaceous or Late Jurassic times, the North China Block (NCB) and the South China Block (SCB) were fully assembled with Eurasia [2–11]. However, noticeable differences still exist among the Late Jurassic and Cretaceous poles reported by different workers for both blocks as well as between the Chinese and Eurasian coeval poles. Various possibilities have been proposed to explain the discrepancies, such as inaccurate ages of magnetization, uncertainties associated with Eurasian APWP, and post-collisional motion of sampling sites [7,8,10–12]. Obviously, more intensive studies have to be undertaken before the different hypotheses can be properly evaluated. In 1986, we reported a provisional Cretaceous/Tertiary paleomagnetic pole from the Upper Cretaceous Xiaoba Formation and lower Tertiary Leidashu Formation from the Huili area of southwestern Sichuan province, which indicated a $13 \pm 10.3°$ northward translation and $22 \pm 11.2°$ counterclockwise rotation of the sampling area relative to Eurasia [2]. However, as we stressed in that paper, the determination was based on only 12 hand samples from two rock formations, and a more intensive study was initiated to verify this result.

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In 1988, we revisited the Huili area. A more extensive paleomagnetic sampling was carried out on both the previously sampled Xiaoba and Leidashu Formations and previously unsampled Jurassic and Lower Cretaceous formations. The Jurassic data have been published elsewhere [10], and this paper reports new Cretaceous and lower Tertiary results.

Geology and sampling

The Huili area is located in the western part of the SCB, with the well-known Sichuan redbed basin to the northeast and the Central Yunnan redbed basin to the southwest (Fig. 1a). The Cretaceous sequence in the area starts with the Lower Cretaceous Feitianshan Formation, which disconformably rests on the Upper Jurassic Guangou Formation and has a localized areal distribution. The Feitianshan Formation is 410 m thick at the section sampled and is characterized by a series of cyclic, lenticular, conglomerate, brick-red sandstone, siltstone and calcareous mudstone deposited in a fluvial environment [13]. The lower part of the formation bears an ostracod assemblage *Minheella–Pinnocypridea*, and the upper part *Cypridea–Latonia* [Leng Zhengwen, pers. commun., 1990]. The Feitianshan Formation probably spans from the Berriasian to Barremian ages [14]. The Xiaoba Formation is divisable into three members. The Lower Member, also called the Datongchang Member, disconformably overlies the Feitianshan Formation or Jurassic rocks, and consists of purplish-red, thick-bedded quartz arkose and copper-bearing conglomerate in the lower part and siltstone and

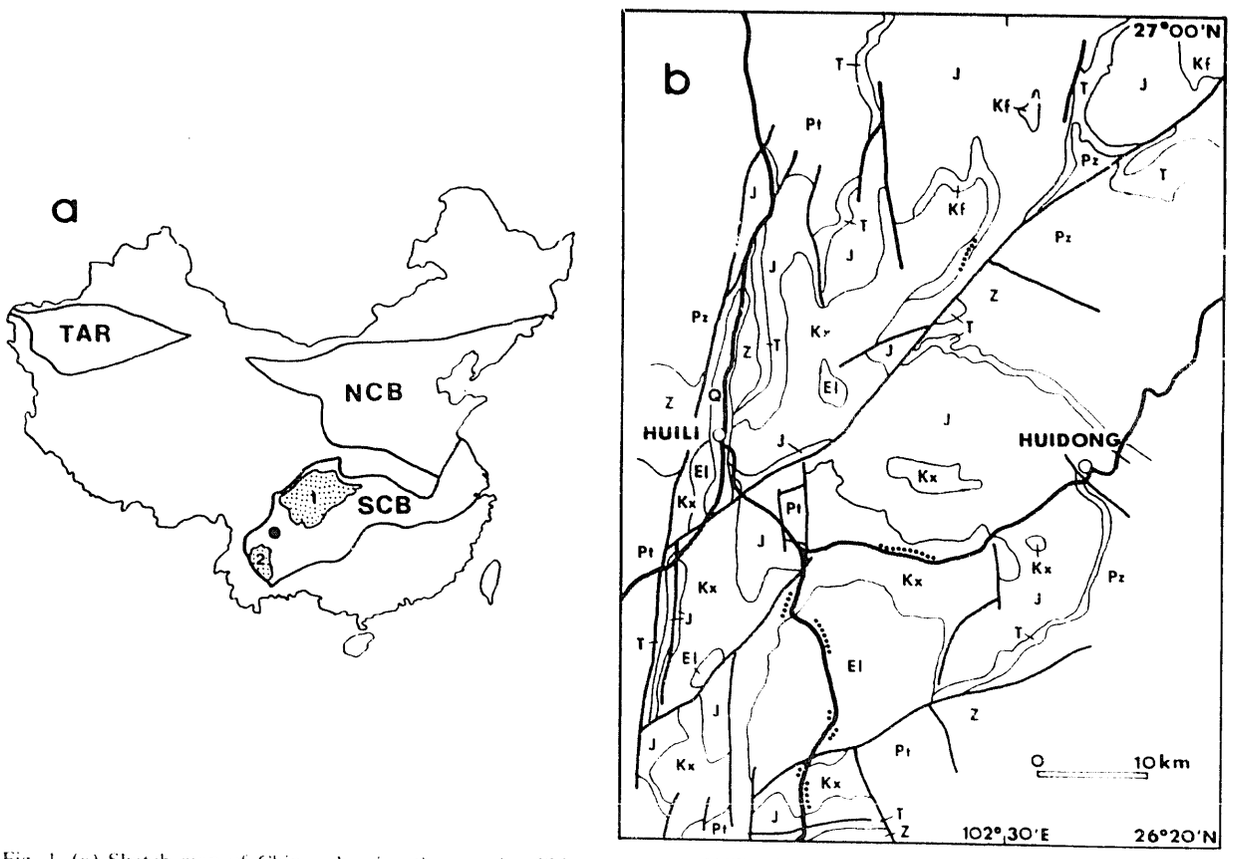


Fig. 1. (a) Sketch map of China, showing three major Chinese cratons: NCB = North China Block; SCB = South China Block; TAR = Tarim block. Stippled area indicates two major Mesozoic redbed basins in western SCB: *1* = Sichuan Basin; *2* = Central Yunnan Basin. Black circle denotes the sampling area of this study. (b) Simplified geologic map of the sampling area. Wide lines are major roads, and medium lines are faults. Dots denote sampling sites. Rock units: Q = Quaternary; El = Paleocene–Eocene Leidashu Fm; Kx = mid–upper Cretaceous Xiaoba Fm; Kf = lower Cretaceous Feitianshan Fm; J = Jurassic; T = Upper Triassic; Pz = Paleozoic; Z = Sinian and Pt = Proterozoic.

mudstone in the middle and upper parts. The Middle Member is composed of purplish-red siltstone and mudstone interbedded with marls. The Upper Member is sometimes missing, and is characterized by bright purplish-red mudstone when present. The Xiaoba Formation yields abundant ostracods, such as *Cypris* sp., *Eucypris* sp. and *Cyprinotus* sp., and charophytes such as *Obtusochara* sp. and *Sphaerochara* sp. [13]. Hao et al. [14] correlate the Datongchang Member with the Jiaguan Formation in the Sichuan basin and assign a middle Cretaceous age to it (roughly the Cenomanian to Turonian ages) according to the three-fold subdivision of the Chinese Cretaceous System they proposed. Consequently, the Middle and Upper Members of the Xiaoba Formation are equivalent to the Guankou Formation in the Sichuan basin and are of Late Cretaceous age (from the Coniacian to Maastrichtian ages). The total thickness of the Xiaoba Formation varies from 850 to 1300 m. The disconformably overlying Leidashu Formation can also be divided into three members, starting with thick-bedded, purplish-grey feldspathic quartz sandstone and lenticular, pebble-bearing sandstone in the Lower Member, going up through purplish-red feldspar-bearing quartz sandstone interbedded with calcareous siltstone in the Middle Member, and ending up with purplish-red feldspar-bearing quartz sandstone, calcareous siltstone and mudstone with marly intercalations in the Upper Member. This formation totals 200–1450 m in thickness [13]. A provisional age of Late Cretaceous to Paleocene was previously assigned to the Leidashu Formation [13]. In recent years, the age of this formation has been changed to Lower Tertiary [14,15]. Hao Yichun [pers. commun., 1991] further suggests that a Paleocene to Eocene age might be more appropriate for the Leidashu Formation because the study of the ostracods from the formation indicates a Paleocene or late Paleocene age and the charophytes a Paleocene to Eocene age. Since the folded redbed sequence in the Huili area is unconformably overlain by the flat-lying Neogene Xigeda Formation, the folding took place after the deposition of the Leidashu Formation.

Our 1988 sampling strategy was different from that of 1986. First, different rock formations were sampled separately and more extensively and, second, rocks of the same formation were collected, where possible, at different flanks of the fold so that a fold test for paleomagnetic stability could be performed. Samples were collected from seven sites where the Feitianshan Formation was exposed in a monocline at a type section (Shuangma) about 55 km northeast of downtown Huili (Fig. 1b), where the rocks dip northwest at 18–29°.

The Xiaoba and Leidashu Formations were sampled from the northern and southern flanks of a syncline southeast of Huili (Fig. 1b). Rocks of the Xiaoba Formation on the northern flank dip south, locally east, at 13–34°; whereas those on the southern flank dip north at 17–25°, but locally at 73° due to faulting. Being in the core of the syncline, the bedding dip of the Leidashu Formation is very gentle, at only 6–20°. Sampling of these two formations was carried out along the highways from Huili to Huidong and from Huili to Tongan. Altogether, samples from twenty-two sites were collected from the Lower and Middle Members of the Xiaoba Formation, and from twelve sites from the Middle and Upper Members of the Leidashu Formation. Five cores 2.54 cm in diameter were usually taken from each site using a portable rock drill and oriented with a magnetic compass.

Laboratory method

Laboratory work was conducted at the University of Florida and thermal cleaning was used throughout the investigation. Stepwise demagnetization was performed in a TSD-1 Schonstedt thermal demagnetizer and samples were usually subjected to between 7 and 21 heating steps at temperatures up to 705°C. Remanence of the samples was measured in an ScT two-axis cryogenic magnetometer. All the equipment was housed in a magnetically shielded room with residual field of 50–250 nT. Demagnetization behavior of individual samples was examined by plotting the demagnetization data on an orthogonal projection [16]. Paleomagnetic directions were calculated using principal component analysis [17] and site mean statistics were computed following Fisher [18]. In computing site or formation means, sample directions with a maximum angular deviation (MAD) or site mean directions with α_{95}

TABLE 1

Site mean directions for the Paleocene–Eocene Leidashu Formation (26.4°N, 102.3°E)

Site	Bedding	n/N	In situ		Tilt corrected		α_{95}	k_2	Pole position	
			D(°)	I(°)	D(°)	I(°)			Lat. (°N)	Long. (°E)
C	230/12	5/5	4.7	27.9	1.1	19.1	5.8	174.5	73.4	278.5
D *	76/6	4/6	200.6	−4.9	201.0	−9.8	17.4	28.8	−60.6	55.6
E	58/11	4/5	180.3	−13.5	182.3	−22.7	8.1	130.7	−75.3	93.4
F *	58/11	5/6	168.9	−12.7	170.2	−22.9	18.5	18.1	−72.8	136.7
G	57/10	5/5	162.9	1.4	163.1	−8.2	7.1	118.6	−62.5	141.2
H	57/10	5/5	170.6	−3.6	171.2	−12.7	9.0	73.1	−68.3	126.6
I	57/10	5/6	189.2	2.8	189.3	−4.6	5.3	207.2	−64.3	80.4
J *	57/10	4/5	178.0	−9.9	179.3	−18.4	18.3	26.1	−73.0	104.7
K	257/20	4/5	27.8	26.1	23.6	10.4	3.6	649.4	59.1	231.4
L	339/20	5/5	155.2	−15.1	160.7	−15.5	3.1	625.8	−64.0	150.5
Formation Mean:		7/12 **	178.2	−12.2	178.7	−13.7	12.1		70.6 [a]	286.1
			$k_1 = 15.1$		$k_2 = 25.7$				$A_{95} = 11.6$	

Bedding = strike/dip; n/N = number of samples or sites in calculation for the means/number of samples or sites measured; D, I = declination and inclination; α_{95}, A_{95} = radius of the 95% confidence circle about the mean direction and pole, respectively; k1, k2 = precision parameters before and after tilt correction.

* Excluded from calculation of the formation mean.

** Sites A and B were omitted because samples from these two sites are weathered, see text for details.

[a] North pole.

TABLE 2

Site mean directions for the middle–upper Cretaceous Xiaoba Formation (26.5°N, 102.4°E)

Site	Bedding	n/N	In situ		Tilt corrected		α_{95}	k_2	Pole position	
			D(°)	I(°)	D(°)	I(°)			Lat. (°N)	Long. (°E)
A	71/17	5/5	355.1	31.8	359.1	48.1	9.9	60.5	87.3	85.4
B	71/17	5/5	4.1	36.2	11.5	51.4	6.7	131.5	78.6	161.2
C	105/16	5/5	5.8	24.1	4.0	39.9	12.8	36.5	84.7	239.0
D	105/16	5/5	359.5	19.5	357.1	34.9	5.6	191.0	82.2	302.6
E *	101/13	5/5	338.9	43.5	330.0	54.0	18.4	18.2	63.1	36.8
F	112/19	6/6	4.2	27.2	359.3	45.1	4.6	214.5	89.4	13.0
G	101/19	5/5	357.0	20.5	354.1	38.6	9.6	64.1	82.8	331.7
H	91/16	5/5	355.3	20.0	354.4	36.0	9.4	67.0	81.6	321.2
I	98/13	5/5	359.9	23.3	358.7	36.1	7.9	93.9	83.3	293.0
J	98/13	5/5	358.8	27.7	357.3	40.5	8.2	88.4	85.7	318.7
K	306/25	5/5	37.5	48.7	37.1	23.7	9.8	62.3	52.5	207.0
L	306/25	5/5	14.4	68.1	25.0	44.0	9.8	62.0	67.6	188.3
M	310/22	5/5	49.8	63.9	45.8	42.1	13.3	34.1	48.8	184.8
N *	301/23	5/5	33.3	59.1	32.5	36.1	20.3	15.1	59.6	197.9
0	208/73	6/6	50.6	39.1	352.3	28.0	6.5	106.7	76.3	315.5
P *	344/34	4/5	185.5	−20.7	193.0	−5.8	18.9	24.5	−63.3	72.3
Q *	344/34	4/5	348.0	30.2	5.1	22.6	19.7	22.8	74.5	263.4
R	23/16	5/5	353.9	18.0	359.6	25.1	12.0	41.7	76.7	283.9
S	20/22	4/5	342.9	22.2	353.0	34.0	7.2	163.8	79.8	323.2
T	28/25	5/7	355.7	23.4	8.4	34.6	9.1	71.3	79.3	234.8
U	265/17	5/5	41.3	46.3	31.8	33.5	9.2	70.6	59.6	201.0
V	265/17	5/5	32.4	56.7	21.7	42.2	5.5	195.4	70.3	193.3
Formation Mean:		18/22	7.3	35.6	8.1	38.8	6.6		81.5	220.9
			$k_1 = 13.0$		$k_2 = 28.8$				$A_{95} = 7.1$	

* Excluded from calculation of the formation mean.

See Table 1 for other explanations.

values larger than 15° were rejected. Site mean directions and associated statistics are summarized in Tables 1–3.

Results

Paleocene to Eocene Leidashu Formation

The natural remanent magnetization (NRM) intensity of the samples from the Leidashu Formation is relatively weak and variable, with a geometric mean of 1.81 mA/m ± 211% (1σ). Samples from sites A and B are weathered and only yielded erratic directions at high temperatures (Fig. 2a). Thus, these two sites were not considered further. Two magnetic components were revealed from the samples of the remaining sites (Figs. 2b and c). A low temperature component (LTC) accounted for an appreciable amount, often 20–65%, of the initial NRM intensities, and was removed below 400°C. The geographic coordinates of this LTC are scattered, but in general northerly and downward, with a mean of D/I = 352.3°/35.8° (α_{95} = 11.6°, k = 4.3, N = 46 samples) which, being close to the present dipole field direction at the sampling locality (I = 44.8°), is most likely a recent overprint. A ChRM direction was isolated usually in the temperature range 600–700°C. Two polarities are present, with eight sites being reversed (southerly and up) (Fig. 2b) and two sites normal (northerly and down) (Fig. 2c), but in either case the inclinations are very shallow. Three of the reversed sites (D, F and J) have larger α_{95} values due to within-site scatter and were excluded from calculation of the formation mean. The tilt-corrected mean direction of the Leidashu Formation based on the remaining seven sites is D/I = 178.7°/ − 13.7° (α_{95} = 12.1°), which corresponds to a north pole at 70.6°N/286.1°E (A_{95} = 11.6°). Despite the very gentle dip of the bedding, it is interesting to note that the clustering of the site mean directions improves after bedding correction, although statistically insignificant, with an increase in k from 15.1 to 25.7 (Table 1 and Fig. 3).

Middle to upper Cretaceous Xiaoba Formation

The NRM intensity of the samples of most sites from the Xiaoba Formation is stronger and more internally consistent than that of the Leidashu Formation, with a geometric mean of 6.31 mA/m ± 150% (1σ). Except for a few samples which showed more-or-less univectorial behavior during demagnetization (Fig. 4b), most samples displayed two magnetization components (Figs. 4a, c, e and f). Nearly all Xiaoba magnetizations are of normal polarity. A LTC, which might also be a recent overprint (D/I = 4.7°/46.0° α_{95} = 10.5°, k = 6.1, N = 37 samples), was eliminated below 400°C but only accounted for 5–20% of the initial NRM intensities. The high temperature characteristic components isolated from the samples from the northern flank of the syncline are northerly with downward inclinations that are shallower than that of the present dipole field before tilt correction, but become steeper after bedding correction is applied (Fig. 3). On the

TABLE 3

Site mean directions for the Lower Cretaceous Feitianshan Formation (26.8°N, 102.5°E)

Site	Bedding	n/N	In situ		Tilt corrected		α_{95}	k_2	Pole position	
			D(°)	I(°)	D(°)	I(°)			Lat. (°N)	Long. (°E)
A	222/29	5/5	39.3	48.3	11.7	39.6	3.3	541.0	78.6	212.2
B	222/29	3/5	43.6	44.4	17.7	38.4	9.1	183.7	73.1	206.6
C	222/29	5/5	46.5	42.2	21.6	38.0	2.8	728.2	69.6	203.5
D	222/29	5/5	49.2	38.9	26.2	36.6	5.8	177.2	65.5	201.9
E	228/24	6/6	46.4	38.0	29.1	33.6	5.6	142.3	62.0	203.9
F	222/18	6/6	39.9	38.5	26.3	35.7	3.1	466.8	64.9	203.2
G	222/18	5/5	35.6	41.2	21.3	36.9	5.9	170.9	69.6	205.8
Formation Mean:		7/7	43.0	41.7	22.1	37.1	3.8		69.0	204.6
			k_1 = 242.8		k_2 = 251.1				A_{95} = 43	

See Table 1 for explanation.

other hand, the ChRM conponents of the samples from the southern flank of the syncline are northeasterly and steeply downward in geographic coordinates, and become shallower upon tilt correction, thus merging with those from the northern flank in stratigraphic coordinates (Fig. 3). Samples from site P are weakly magnetized, with initial NRM intensities ranging from 0.57 to 1.49 mA/m, comparable to that of the Leidashu Formation. These sample magnetizations showed a reversed (southerly and up) but very unstable direction (Fig. 4d). As a result, the mean direction of this site yielded a large α_{95} value (18.9°) and was excluded from computation of the formation mean. Sites E, N and Q of normal polarity were also excluded from computation of the formation mean owing to large within-site scatter ($\alpha_{95} > 18°$, Table 2). The tilt-corrected mean direction for the Xiaoba Formation based on the remaining eighteen normal polarity site means is $D/I = 8.1°/38.8°$ ($\alpha_{95} = 6.6°$), corresponding to a

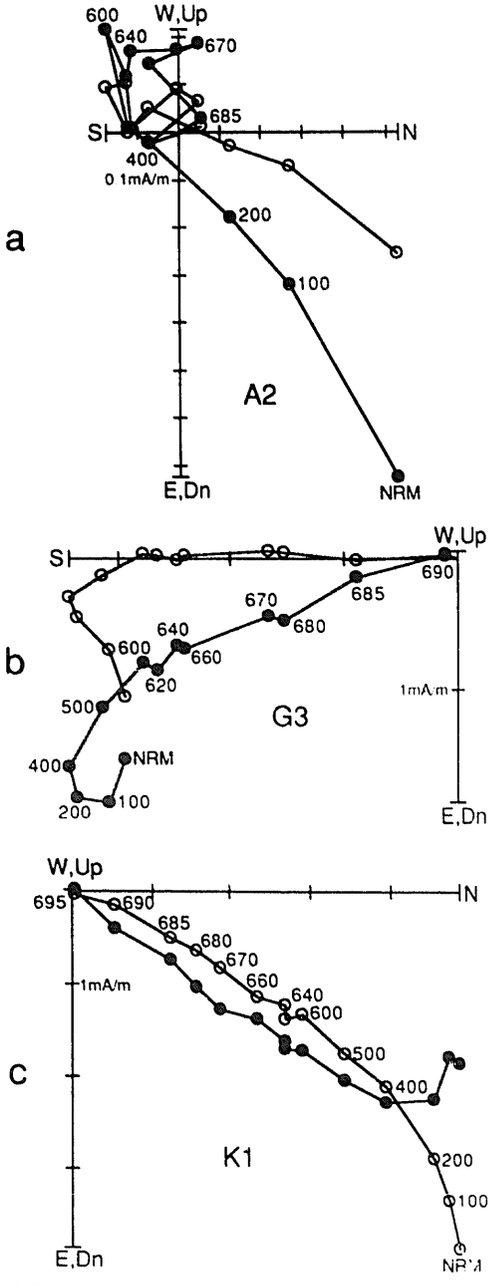


Fig. 2. Representative orthogonal plots of thermal demagnetization (in geographic coordinates) for Leidashu Formation. Solid (open) circles are plotted on horizontal (vertical) planes. Thermal treatment levels are in degrees Celsius.

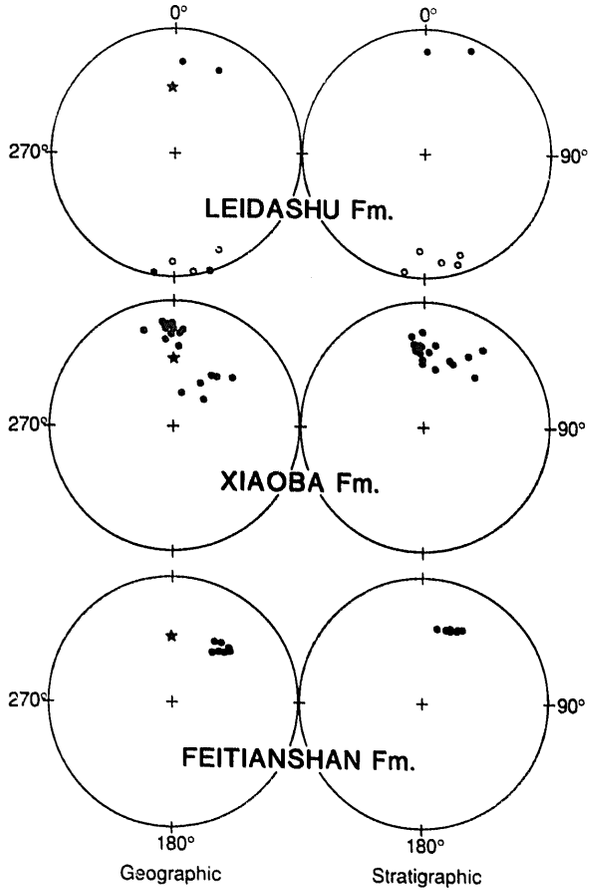


Fig. 3. Equal-area projections of site mean directions before and after tilt correction for Leidashu, Xiaoba and Feitianshan formations. Solid (open) circles represent directions plotted onto the lower (upper) hemisphere. Star denotes the present dipole field direction at the sampling locality.

paleopole at 81.5°N/220.9°E ($A_{95} = 7.1°$). This direction passes the fold test at the 95% confidence level ($k_2/k_1 = 2.22 > 1.78$, test value from [19]).

Lower Cretaceous Feitianshan Formation

Samples from the Feitianshan Formation are as strongly and consistently magnetized as most

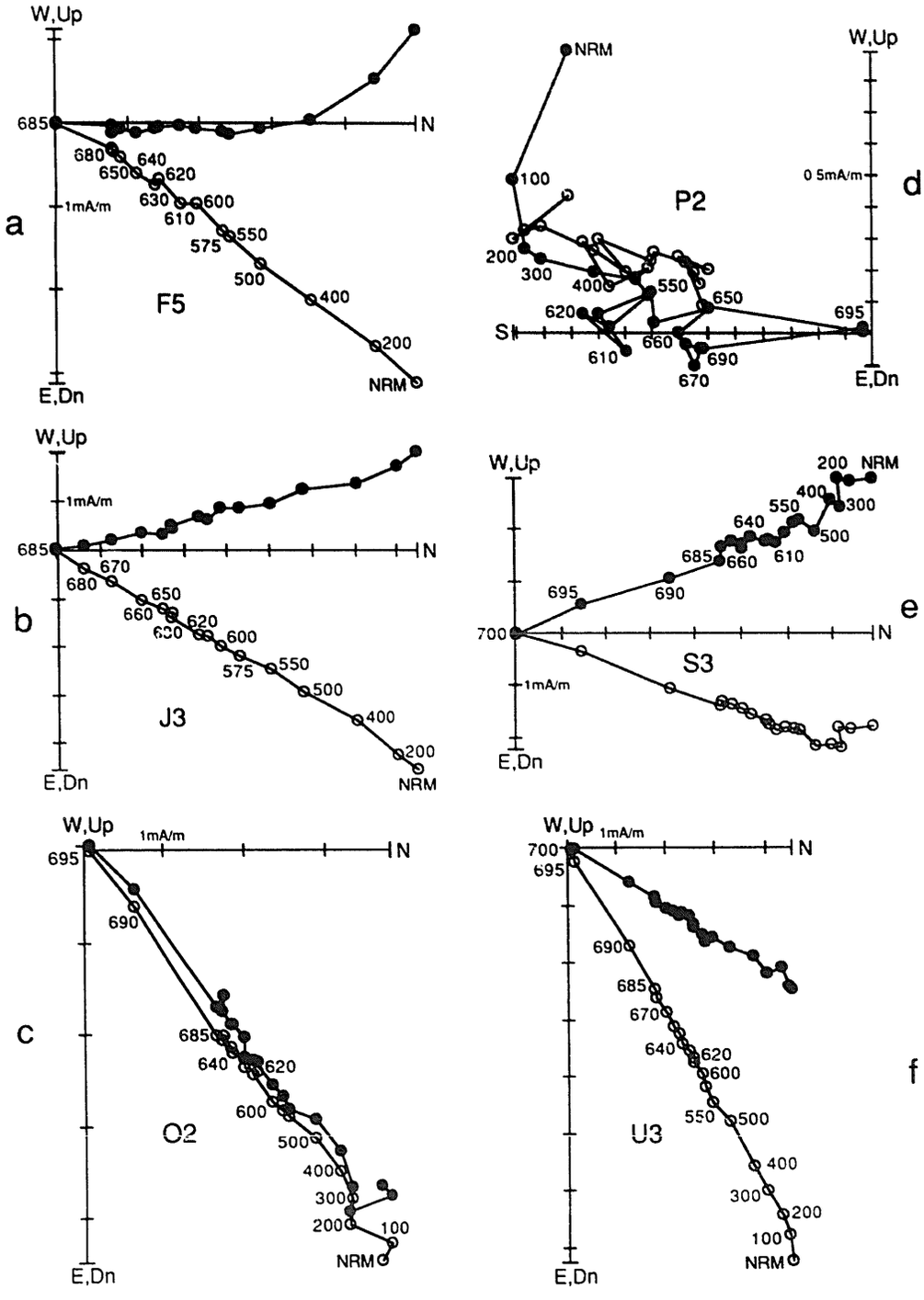


Fig. 4. Representative orthogonal plots of thermal demagnetization for Xiaoba Formation. Coordinates and symbols as in Fig. 2.

of the Xiaoba Formation, with a geometric mean of 7.79 mA/m ± 143% (1σ) for the NRM intensities. Unlike the two formations described above, most samples from this formation univectorially decayed with increasing demagnetization temperatures (Fig. 5c). Some samples from sites A, B and C possessed two components: a LTC was apparently oriented in the present-day geomagnetic field direction and a high temperature ChRM component was resolved at above 400°C and is of the same direction as those showing univectorial behavior (Fig. 5b). The ChRM directions in both cases are northeasterly and moderately down and deviate from the present-day geomagnetic field in geographic coordinates. Two samples from site B contain small pebbles. Their NRM was dominated by a large LTC and the high temperature directions are quite different from that of other samples (Fig. 5a), probably because of the pebbles. These two samples were excluded from computation of the site mean. The tilt-corrected mean direction of the Feitianshan Formation is $D/I = 22.1°/37.1°$ ($\alpha_{95} = 3.8°$) (Table 3 and Fig. 3), corresponding to a paleopole at 69.0°N/204.6°E ($A_{95} = 4.3°$).

Discussion

The ChRM directions isolated from the Paleocene–Eocene Leidashu Formation are substantially shallower than the present dipole field direction and possess two polarities. The grouping of the site mean directions is improved after tilt correction despite very gentle bedding dips. It is safe to say, therefore, that the ChRM of the Leidashu Formation is an early acquired magnetization. Twenty-one of the 22 sites sampled from the Xiaoba Formation are of normal polarity. The fold test is positive for the mean ChRM direction of the normal sites and constrains the age of remanence acquisition to be prefolding, probably pre-Eocene. Although site P exhibits reversed directions, the weak and unstable magnetization of this site discredits the reliability of these reversed directions. The prevalence of normal polarity in the sampled Mid–Lower members of the Xiaoba Formation suggests that the ChRM of these rocks was probably acquired during the Cretaceous Long Normal Superchron [20]. The mean ChRM direction for the Lower Cretaceous Feitianshan Formation is based upon seven sites from a monocline, but its deviation from the present in situ dipole field, as well as good internal consistency, speaks for early acquisition of the remanent magnetization.

The paleolatitude for the Huili area indicated by the mean inclination of the Paleocene–Eocene Leidashu Formation is 6.9 ± 6.3°N, far lower than the value of 34.3 ± 4.3°N predicted by the Eurasian APWP [21], if an age of 50 Ma is

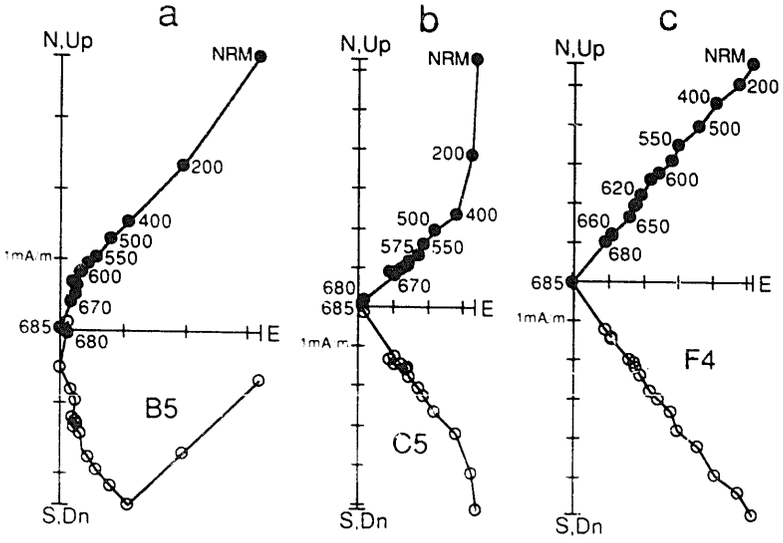


Fig. 5. Representative orthogonal plots of thermal demagnetization for Feitianshan Formation. Coordinates and symbols as in Fig. 2.

assumed for the magnetization. Consequently, the corresponding paleopole is far-sided with respect to the Eurasian mean Eocene pole (Fig. 6a). If the SCB was fully assembled with Eurasia by Cretaceous times, as discussed below, the difference between the observed and predicted paleolatitudes would imply an intracontinental crustal shortening of as large as 3040 ± 840 km north of the sampling area since the early Tertiary. This is not supported by geological evidence [cf. 7].

The cause for this anomaly is not well understood. Anomalously low Eocene paleolatitudes were noted from some other regions of the world, for instance, the western Tethyan region (the Southern Alps, Taurides, Iran, Afghanistan and Greece), where the observed Eocene paleolatitudes are 5–15° lower than the predicted values even from the African APWP [see 22], R.J. Enkin, pers. commun., 1992]. The available paleomagnetic data, however, are not sufficient to demonstrate that this is due to abnormal behavior of the Earth's magnetic field.

That the sediments were deposited on a dipping surface is a second possibility. If there existed any initial dip during the sedimentation of the Leidashu Formation, it would be more likely to occur in the coarse-grained Lower Member of the formation. Moreover, combining results from different sites should have reduced any systematic directional bias if an initial dip were present at a single site [23]. Substantial inclination shallowing has been observed in both naturally and experimentally deposited sediments, and attributed to misalignment of magnetic particles during deposition or post-depositional processes [24–30]. For example, Tauxe and Kent [28] found that the modern sediments of the Soan River in northern Pakistan correctly recorded the declination of the Earth's field but that the recorded inclination was 25° shallower. The Jurassic to lower Tertiary redbeds in the Huili area are fluviolacustrine deposits [13]. It is possible that the observed inclination shallowing of the Leidashu Formation is an inclination error related to depositional or post-depositional processes. Further questions, however, arise, such as: is the remanence carried predominantly by detrital hematite and why did inclination errors only occur in the Leidashu Formation? These questions cannot be answered without further intensive rock magnetism studies. Nevertheless, it appears that the inclination shallowing of the Leidashu Formation is a contributing factor to the shallow inclination of the provisional mean direction we previously

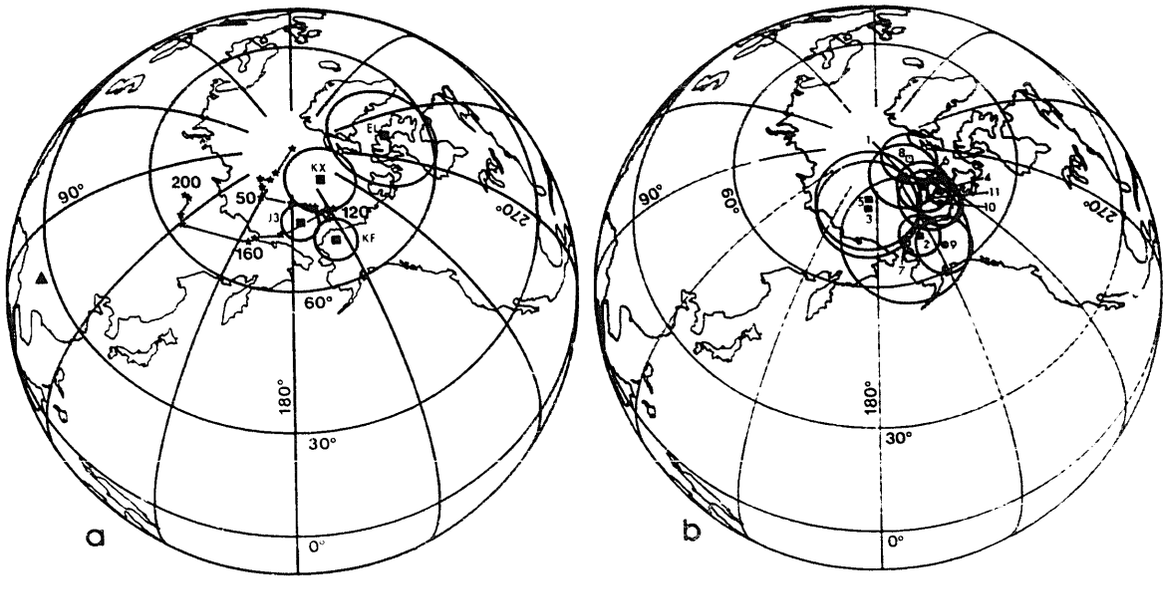


Fig. 6. (a) Paleomagnetic poles determined from Jurassic to lower Tertiary rock formations from the Huili area. EL = Leidashu Fm; KX = Xiaoba Fm; KF = Feitianshan Fm; J3 = Upper Jurassic. Triangle denotes the sampling area. The Eurasian APWP [21] is marked with stars. (b) Cretaceous paleopoles for Chinese blocks. Numbers denoting the poles are the same as in Table 4.

reported, which included at least two samples of very shallow inclinations from the Leidashu Formation [2].

The mean direction for the Xiaoba Formation is not significantly different from that previously determined for the Xiaoba and Leidashu formations [2], with an angular separation of 11.2 ± 15.2° between the two. Inasmuch as the newly determined mean direction is derived from eighteen sites (91 samples) and does not appear to have anomalously shallow inclinations, we believe that it is a more reliable determination of the mid–late Cretaceous magnetic field direction for the study area. The corresponding pole position is coincident with other Cretaceous poles reported for the Chinese blocks and Eurasia (Fig. 6b), thus reinforcing the assertion that, by Cretaceous times, the Chinese blocks had been fully assembled with Eurasia [2,7,8,11]. Using the 100 Ma mean pole of the Eurasian APWP proposed by Besse and Courtillot [21], the expected declination and paleolatitude for the Huili area are 14.6 ± 5.5° and 24.7 ± 5.0°N, respectively. Comparison of these with the declination and paleolatitude observed from the Xiaoba Formation indicates neither significant rotation (6.5 ± 10.1°) nor latitudinal translation (2.8 ± 6.9°) of the Huili area relative to Eurasia. This is consistent with the Jurassic results from the same area [10].

Our 1988 paleomagnetic sampling involved all formations of the Jurassic to lower Tertiary sequence of the Huili area. To our satisfaction, most of the sampled formations have yielded good results. Although our investigation was not designed with magnetostratigraphy in mind, the proportions of the accepted normal and reversed sites do seem to fit broadly with the established Jurassic to lower Tertiary reversal pattern from geomagnetic time scales [e.g., 20]. The polarities appear to be mixed in the Lower Jurassic Yimen Formation (reversed: normal sites = 4: 4, see [10]). The Upper Jurassic Niugundang and Guangou formations are dominated by normal polarity (16 normal sites out of 20, see [10]). While no reliable reversed site is seen in the sampled Cretaceous Feitianshan and Xiaoba formations, the reversed field appears to be dominant in the Paleocene–Eocene Leidashu Formation (Tables 1–3). Proportions of reversed/normal sites are, of course, not a substitute for magnetic polarity stratigraphy, yet the general pattern suggests that the paleontologically assigned ages for the formations sampled are not in serious error.

A prominent hairpin turn features Besse and Courtillot's Eurasian APWP, which moves rapidly along the track from 170 Ma to 130 Ma, then stalls at 130–80 Ma, and moves rapidly again from 80 Ma to 50 Ma [21]. In Southeast Asia this hairpin turn would be seen primarily as variations in the inclination of remanence [7]. Specifically, if the SCB was assembled with Eurasia during Jurassic times [3–6,9,10, 31], the inclination of remanence of the Cretaceous rocks from the SCB should be shallower than that of the Upper Juras-

TABLE 4

Cretaceous poles for the Chinese blocks used in this paper

No.	Location	Rock age	Mean pole	A_{95}	Reference
1	SCB, SW Sichuan	K_{2-3}	81.5°N/220.9°E	7.1°	This study (Table 2)
2	SCB, SW Sichuan	K_1	69.0°N/204.6°E	4.3°	This study (Table 3)
3	SCB, Nanjing	K_3	76.3°N/172.6°E	10.3°	[2]
4	SCB, Sichuan Basin	J_3–E	73.9°N/234.4°E	2.9° *	[8]
5	SCB, Hong Kong	K_3	78.2°N/171.9°E	10.6°	[11]
6	SCB, Zhejiang	K_1	77.1°N/227.6°E	5.5°	[32]
7	NCB, Shandong	K_1	69.1°N/197.0°E	13.2°	Recalculated from [32]
8	NCB, Inner Mongolia	K	82.9°N/249.5°E	5.7°	[33]
9	Tarim	J–K	65.0°N/214.0°E	6.0°	[34]
10	Junggar	J/K	72.3°N/227.3°E	4.8°	[35]
11	Junaaar	K/T	74.3°N/223.1°E	6.4°	[35]

No. = Numbers denoting palcopoles in Fig. 6b.

* Estimated from original dm and dp values.

sic rocks. This is consistent with the Cretaceous results presented here and Jurassic results reported earlier [10].

Finally, some noticeable differences still exist among reported Cretaceous poles for the Chinese blocks (Fig. 6b and Table 4). The uncertainties associated with the age of the rock and age of the magnetization are likely contributing factors to this, as well as relative rotation of the sampling localities. Part of these differences can be reduced when more intensive investigations are conducted, as demonstrated in this study. The deviation of the Lower Cretaceous pole from the track between the Upper Jurassic and Upper Cretaceous poles (Fig. 6a) might be eliminated by sampling a different locality or two limbs of a fold. Resolution of the poles may also be improved if resolution of rock age can be improved and/or age of the remanence can be better constrained in some of the earlier studies.

Conclusion

A re-investigation of the Cretaceous to Lower Tertiary rocks from southwestern Sichuan province of China reveals anomalously shallow inclinations from the Paleocene–Eocene Leidashu Formation, which might be related to inclination error associated with depositional and/or post-depositional processess. The paleopole derived from the mid–upper Cretaceous Xiaoba Formation, based upon a larger sample size and apparently free from effects of inclination shallowing as suggested for the Leidashu Formation, is coincident with other Cretaceous poles for the Chinese blocks and Eurasia. This reinforces the assertion that, by Cretaceous times, the Chinese blocks had been fully assembled with Eurasia.

This study demonstrates that minor differences among the reported Cretaceous poles for the Chinese blocks can be reduced if more extensive sampling is undertaken and if resolution of rock and magnetization age can be improved in some of the formations studied earlier.

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References

1 M.W. McElhinny, B.J.J. Embleton, X.H. Ma and Z.K. Zhang, Fragmentation of Asia in the Permian, Nature 293, 212–216, 1981.

2 D.V. Kent, G. Xu, K. Huang, W.Y. Zhang and N.D. Opdyke, Paleomagnetism of upper Cretaceous rocks from South China, Earth Planet. Sci. Lett. 79, 179–184, 1986.

3 J.L. Lin, M. Fuller and W.Y. Zhang, Preliminary Phanerozoic polar wander paths for the North and South China blocks, Nature 313, 444–449, 1985.

4 N.D. Opdyke, K. Huang, G. Xu, W.Y. Zhang and D.V. Kent, Paleomagnetic results from the Triassic of the Yangtze platform, J. Geophys. Res. 91, 9553–9568, 1986.

5 X. Zhao and R.S. Coe, Tectonic implication of Permo-Triassic paleomagnetic results from North and South China, in: Deep Structure and Past Kinematics of Accreted Terranes, J.W. Hillhouse, ed., IUGG Geophys. Monogr. 5, 267–283, 1989.

6 M. Steiner, J. Ogg, Z. Zhang and S. Sun, The Late Permian/Early Triassic magnetic polarity time scale and plate motions of South China, J. Geophys. Res. 94, 7343–7363, 1989.

7 R.J. Enkin, Y. Chen, V. Courtillot, J. Besse, L. Xing, Z. Zhang, Z. Zhuang and J. Zhang, A Cretaceous pole from South China, and the Mesozoic hairpin turn of the Eurasian apparent polar wander path, J. Geophys. Res. 96, 4007–4028, 1991.

8 R.J. Enkin, V. Courtillot, L. Xing, Z. Zhang, Z. Zhuang and J. Zhang, The stationary Cretaceous paleomagnetic pole of Sichuan (South China Block), Tectonics 10, 547–559, 1991.

9 Z. Yang, X. Ma, J. Besse, V. Courtillot, L. Xing, S. Xu and J. Zhang, Paleomagnetic results from Triassic sections in the Ordos Basin, North China, Earth Planet. Sci. Lett. 104, 258–277, 1991.

10 K.N. Huang and N.D. Opdyke, Paleomagnetism of Jurassic rocks from southwestern Sichuan and the timing of the closure of the Qinling Suture, Tectonophysics 200, 299–316, 1992.

11 L.S. Chan, Paleomagnetism of late Mesozoic granitic intrusions in Hong Kong: implications for Upper Cretaceous

reference pole of South China, J. Geophys. Res. 96, 327–335, 1991.
12 K.H. Kim and R. Van der Voo, Jurassic and Triassic paleomagnetism of South Korea, Tectonics 9, 699–718, 1990.
13 Sichuan Regional Stratigraphic Tables Compilation Team, Regional stratigraphic tables of southwest China (Sichuan Province), 672 pp., Geol. Publishing House, Beijing, 1978 (in Chinese).
14 Y. Hao and thirteen others, The Cretaceous System of China, 301 pp., Geol. Publ. House, Beijing, 1986 (in Chinese with English abstract).
15 Y. Li and ten others, The Tertiary System of China, 300 pp., Geol. Publ. House, Beijing, 1986 (in Chinese with English abstract).
16 J.D.A. Zijderveld, A.C. demagnetisation of rocks: analysis of results, in: Methods in Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., pp. 254–286, Elsevier, Amsterdam, 1967.
17 J.L. Kirschvink, The least-squares line and plane and the analysis of palaeomagnetic data, Geophys. J.R. Astron. Soc. 62, 699–718, 1980.
18 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London Ser. A 217, 295–305, 1953.
19 M.W. McElhinny, Statistical significance of the fold test in palaeomagnetism, Geophys. J.R. Astron. Soc. 8, 338–340, 1964.
20 W.B. Harland, R.L. Armstrong, A.V. Cox, L. Craig, A.G. Smith and D.G. Smith, A Geologic Time Scale 1989, 263 pp., Cambridge University Press, Cambridge, 1990.
21 J. Besse and V. Courtillot, Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma, J. Geophys. Res. 96, 4029–4050, 1991.
22 M. Westphal, M.L. Bazhenov, J.P. Lauer, D.M. Pechersky and J.C. Sibuet, Paleomagnetic implications of the evolution of the Tethys belt from the Atlantic Ocean to the Pamirs since the Triassic, Tectonophysics 123, 37–82, 1986.
23 P. Swenson and M. McWilliams, Paleomagnetic results from the Upper Cretaceous Maudlow and Livingston Formations, Southwest Montana, Geophys. Res. Lett. 16, 669–672, 1989.
24 E.A. Johnson, T. Murphy and O.W. Torreson, Pre-history of the earth's magnetic field, Terr. Magn. Atmos. Elec. 53, 349–372, 1948.
25 R.F. King, Remanent magnetism of artificially deposited sediments, Mon. Notic. R. Astron. Soc. Geophys. Suppl. 7, 115–134, 1955.
26 D.H. Griffiths, R.F. King, A.I. Ress and A.E. Wright, Remanent magnetism of some recent varved sediments, Proc. R. Soc. London Ser. A 256, 359–383, 1960.
27 R.A. Blow and N. Hamilton, Effect of compaction on the acquisition of a detrital remanent magnetization in fine grained sediments, Geophys. J.R. Astron. Soc. 52, 13–23, 1978.
28 L. Tauxe and D.V. Kent, Properties of a detrital remanence carried by haematite from study of modern river deposits and laboratory redeposition experiments, Geophys. J.R. Astron. Soc. 77, 543–561, 1984.
29 S. Levi and S. Banerjee, On the origin of inclination shallowing in redeposited sediments, J. Geophys, Res. 95, 4383–4389, 1990.
30 P. Arason and S. Levi, Compaction and inclination shallowing in deep-sea sediments from the Pacific Ocean, J. Geophys. Res. 95, 4501–4510, 1990.
31 R.J. Enkin, Formation et deformation de l'Asie depuis la fin de l'Ere Primaire, 333 pp., Ph.D. Thesis, Univ. Paris VII, 1990.
32 J.L. Lin, The apparent polar wander paths for the North and South China Blocks, 248 pp., Ph.D. Thesis, Univ. California, Santa Barbara, 1984.
33 X. Zhao, R.S. Coe, H. Wu and J. Wang, New paleomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan, Tectonophysics 181, 43–81, 1990.
34 Y. Li, Z. Zhang, M. McWilliams, R. Sharps, Y. Zhai, Y.A. Li, Q. Li and A. Cox, Mesozoic paleomagnetic results of the Tarim craton: Tertiary relative motion between China and Siberia? Geophys. Res. Lett. 15, 217–220, 1988.
35 Y. Chen and ten others, Paleomagnetic study of Mesozoic continental sediments along the Northern Tien Shan (China) and heterogeneous strain in central Asia, J. Geophys. Res. 96, 4065–4082, 1991.